MaternaSync

Future Implementations

A consolidated feature roadmap for Doctor, Patient, Care Team, and Clinical Guidelines.

Product direction
Connect the maternal-care journey across records, patient questions, clinical documentation, care-team tasks, and trusted reference material — while keeping clinical decisions and approvals with authorized professionals.

Planning note

The existing-feature inventory below is based on the supplied project notes and code/navigation review; it has not been independently verified end to end. Features in this document are proposed future implementations, not claims that they already exist.

Prepared for product planning and hackathon prioritization

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1. Scope and existing capabilities

The following capabilities were reported as already present in the code/navigation. They have not yet been tested end to end, so they should be treated as an inventory to validate rather than a verified completion checklist.

Reported existing capability	Primary area
Voice consultation with an AI-generated draft for doctor approval	Doctor
Today’s Brief	Doctor
Document upload with extraction	Shared / clinical workflow
Pregnancy journey timeline	Shared / patient journey
Patient-question workflow with approved replies	Patient + Doctor / Care Team
Tasks, handover and referrals	Care Team + Doctor
Audit log	Governance

This roadmap skips rebuilding those features and focuses on proposed gaps, enhancements, integrations, and the Clinical Guidelines tab.

Product principle

MaternaSync should not become a collection of disconnected AI tools. Its differentiator is the continuity layer: a patient question, report, referral, task, response, owner, and handover should remain linked to the same maternal-care journey.

2. Doctor login — proposed features

Primary user: obstetrician / gynaecologist. The doctor workspace should reduce administrative effort while keeping all clinical decisions, prescriptions, and patient-facing communication under authorized clinician control.

Feature	Product inspiration	MaternaSync implementation idea	Planning priority
Anomaly flags and 30-day trends in Today's Brief	Aayi	Display recorded BP, weight, symptoms, kick-count entries, and adherence logs as dated trends. Flag missing or unusual values using explicitly configured rules; show source, timestamp, and context.	High — only after validation
Risk stratification and next-best-action	Nivrit	If pursued, use a clinician-approved, validated protocol to identify predefined workflow categories (for example, an overdue scan). Show the rule/source and require clinician confirmation. Do not launch unvalidated AI risk scoring.	Defer / safety review
Care pathways per condition	Nivrit	Configurable, versioned pathways for conditions such as GDM, pre-eclampsia, and anaemia, authored and approved by clinical governance. Generate administrative tasks and follow-up reminders from the selected pathway.	High after clinical governance
Prescription writer with interaction alerts	Vaidhji, Gynovate	Create a structured prescription draft from clinician-entered/approved information. Drug, dose, and interaction checking require a verified drug database and validated clinical software; no autonomous dose suggestions.	Defer / high-risk
Hinglish and mixed-language voice capture	Ekko, Vaidhji	Improve transcription for Hindi, Tamil, and English mixed within a sentence; retain the transcript and let the clinician correct terminology before saving.	High
One-tap WhatsApp sharing	Vaidhji, Gynovate	Offer a controlled sharing workflow for approved summaries or documents, with recipient confirmation, consent, delivery status, and minimum necessary information.	Medium; privacy review
Patient explanation generator	Gynovate	Turn an approved note into plain-language counselling text. Label as a draft and require doctor approval before patient release.	High
Document generators	Gynovate	Generate editable drafts for discharge summaries, referral letters, consent forms, medical certificates, and operative notes from verified episode data. Require appropriate professional review/signature.	High
Guideline-aware suggestions with citations	Gynovate	Provide source-attributed reference passages from approved guidelines. Show issuer, title, version/date, page/section, and link; do not convert retrieved text into autonomous treatment advice.	Medium; build reference library first
Auto ICD-10 coding	Vaidhji	Suggest candidate codes from an approved note for clinician/coder review. Preserve the original text and record the final human-selected code.	Medium
Custom note templates	Vaidhji	Let clinicians configure SOAP, ANC visit, follow-up, and handover templates, with required fields and organization-level defaults.	High

Doctor workspace enhancements from the earlier feature proposal

Keep the existing Today's Brief, voice-draft approval, and timeline. Improve them with visible source links, patient-question context, pending-task ownership, configurable note layouts, approved plain-language explanations, and structured document drafts.

3. Patient login — proposed features

Primary user: pregnant or postpartum patient. Patient features should make it easier to communicate with the care team and follow the plan that the care team has already documented.

Feature	Product inspiration	MaternaSync implementation idea	Planning priority
Daily logging in under 2 minutes	Aayi	A short optional check-in for patient-reported symptoms, vitals, mood, and medicine adherence, with timestamps and units. Make it easy to skip and correct entries.	High after safety/privacy review
Kick counter	Aayi	A simple third-trimester movement log, with clinician-approved instructions and clear guidance on contacting the maternity team if concerned. Do not interpret counts as a diagnosis or reassurance.	Defer until clinical review
Medicine and vaccine reminders	Aayi	Reminders based on schedules entered or approved by the care team. Let the patient mark done, skipped, or needs help; avoid independently generating a regimen.	High
AI companion in regional languages	Aayi, Newmi	Support Tamil, Hindi, and Telugu for navigation and general, approved educational information. Medical questions should route into the existing clinician-reviewed workflow; no diagnosis or treatment recommendations.	Medium
Lab report explainer	Aayi	Create a plain-language draft from verified extracted text, with prominent draft status and clinician review before it is shown as an interpretation. Preserve the original report and source.	Defer until validation
Consent controls	Aayi, Vaidhji	Let patients see and manage applicable sharing permissions, recipient, purpose, and duration where supported. Include revocation and a record of consent changes. Align with applicable law and organizational policy.	High / foundational
Trimester content	Newmi, Gynovate	Clinician-approved educational modules for nutrition, safe movement, labour preparation, birth-plan discussions, postpartum recovery, and lactation, with source and review date.	Medium
Visit summary in plain language	Aayi	Generate a patient-friendly version of an approved visit summary and save it to the appointment. Clearly distinguish confirmed instructions from general education.	High

Patient features from the earlier proposal

Retain the existing patient-question workflow and approved replies. Extend it with question status visibility, report/document access, appointment reminders, patient-visible approved instructions, and an accessible pregnancy journey view.

4. Care Team login — proposed features

Primary users: nurses, midwives, coordinators, and authorized administrative staff. These are MaternaSync adaptations of the product ideas; they are not claims that the referenced products offer the same staff-side functions.

Feature	Product inspiration	MaternaSync implementation idea	Planning priority
Compliance tracker and missed-log alerts	Aayi	A work queue for missing patient check-ins, overdue patient-reported logs, and follow-up that the care team has explicitly agreed to perform. Distinguish missing data from a clinical concern.	Medium
Multi-doctor and shared-clinic dashboard	Aayi, Vaidhji	A caseload view across authorized clinicians, showing workload, upcoming visits, unassigned work, and overdue administrative tasks.	High
Pathway-driven task generation	Nivrit	Generate operational tasks from a clinician-approved pathway, such as arranging a scheduled scan. Use versioned rules and avoid automatically issuing clinical orders.	High after pathway governance
Outcome tracking	Nivrit	Aggregate operational metrics such as on-time scans, delivery outcomes where legitimately recorded, and no-show rates. Define data quality, access, and reporting rules before rollout.	Medium
Draft patient replies in the patient’s language	Gynovate, Aayi	Draft administrative or clinician-approved educational responses in the patient's selected language; staff review and route clinical questions to an authorized clinician.	High
WhatsApp reminders and outreach	Vaidhji	Send approved appointment, vaccine, and follow-up reminders from an authorized account, with consent, recipient verification, delivery status, and opt-out handling.	Medium; privacy review
Insurance and certificate paperwork drafts	Gynovate	Prepare structured claims or certificate drafts from verified records for review and signature by the authorized professional.	Medium

Care Team features from the earlier proposal

Keep the current tasks, handovers, and referrals. Strengthen them with owner and due-date visibility, queue filters, reassignment history, shared-clinic workload views, and automatic administrative tasks from approved pathways.

5. Clinical Guidelines tab — Clinical Reference Library

Purpose: give authorized clinicians a source-attributed way to find and review official guidance. This is a reference and retrieval feature, not a clinical decision engine. It should be kept separate from patient-facing AI drafting and must not autonomously recommend treatment.

Recommended source groups

Guideline source	Scope noted for MaternaSync	Implementation note
FOGSI–ICOG GCPRs	Indian obstetric practice; topics may include antenatal care, hypertensive disorders, gestational diabetes, postpartum haemorrhage, induction, multifetal pregnancy, stillbirth, and obesity.	Start here; curate specific documents and confirm current versions.
ICMR Standard Treatment Workflows	Practical Indian workflows, including antenatal diagnosis/management and postpartum haemorrhage.	Start here; verify document title, version, and applicable setting.
MoHFW / Government of India	National maternal-health programmes and public-sector protocols.	Add approved national programme materials relevant to target facilities.
WHO	International guidance on antenatal, intrapartum, postnatal care, and maternal complications.	Add selected documents with publication/update dates.
FIGO	International recommendations, obstetric emergencies, and maternal/fetal medicine.	Add selected, clinically governed references.
RCOG	UK evidence-based obstetric guidance.	Use as a supplementary source; clearly identify jurisdiction.
ACOG / SMFM	US guidance, including high-risk pregnancy and maternal-fetal medicine.	Use as a supplementary source; do not assume direct applicability to Indian practice.
NICE	Clinical pathways covering pregnancy, labour, hypertension, diabetes, and induction.	Use as a supplementary source; identify the issuing body and jurisdiction.
RANZCOG	Australian/New Zealand obstetric and gynaecological guidance.	Use as a supplementary source; identify jurisdiction.

Proposed tab layout

Search and filters: topic, issuing organization, document type, publication/update date, and jurisdiction.
Reference result: title, issuing body, version/date, section/page, short retrieved passage, and original source link.
Document view: readable source content or permitted excerpt, with document metadata and version status.
Attach to workflow: allow a clinician to attach a reference to a note, handover, or internal discussion without automatically injecting its advice into the patient record.
Governance: show review status, last checked date, and whether a document has been superseded.

Suggested technical implementation

1. Curate permitted official documents and preserve their original files.
2. Extract text from PDFs using a parser such as PyMuPDF; use OCR only for scanned pages.
3. Store document metadata, extracted text, and version information in PostgreSQL.
4. Add pgvector semantic retrieval as an optional search layer, while preserving keyword search and source filters.
5. If a local LLM is used to summarize retrieved passages, constrain it to those passages, require source references, label the output as AI-generated, and require clinician review.
6. Keep retrieval logs and document-version history; never present a generated answer without its supporting source.

Source and licensing caution

Use official issuer websites and confirm each document's current version, usage rights, redistribution permissions, and clinical applicability before including it. Do not assume every guideline can be copied into the product or that a search result is current.

6. How the three logins connect

The continuity layer should connect the workflow rather than create three separate data silos.

Stage	Action	Where
1. Patient submits	Question, daily log, uploaded document, or appointment request.	Patient portal
2. System links	Attach the item to the correct patient journey and preserve source, timestamp, and permissions.	Continuity layer
3. Care Team triages	Assign an owner, status, and due date for operational follow-up; route clinical questions to an authorized clinician.	Care Team
4. Doctor reviews	See relevant open items in Today’s Brief; review records and create/approve documentation or a response.	Doctor
5. Patient receives	Show the approved response, visit summary, or reminder according to access and consent settings.	Patient portal
6. Audit and handover	Record relevant changes and carry unresolved items into the next handover or visit.	Shared workflow

Example scenario

A patient submits a question in Tamil and uploads a report. MaternaSync links both to the patient's journey. The care team assigns the question to an authorized clinician. The doctor sees the unresolved question in Today’s Brief, reviews the available source documents, and approves a response. The patient sees the approved response, while the team can track closure and the audit log records the workflow.

Design rule

Every generated clinical or patient-facing output should have a clear source, draft/approved status, named reviewer where appropriate, and a traceable change history. Role permissions must be enforced by the backend, not just hidden UI controls.

7. Recommended implementation priorities

Prioritize features that strengthen the existing product without expanding into unvalidated clinical decision-making.

Priority	Scope
P0 — Validate what exists	Test the current voice-draft approval, Today's Brief, document extraction, timeline, question workflow, tasks/referrals/handovers, and audit log end to end using synthetic data.
P1 — Close continuity gaps	Add clearer ownership and due dates, source links in briefs, queue filters, custom note templates, patient-visible question status, and a plain-language visit-summary draft.
P2 — Language and communication	Improve Tamil/Hindi/English mixed-language transcription, multilingual approved drafts, and appointment reminders with consent and delivery tracking.
P3 — Clinical Reference Library	Start with a small, curated FOGSI–ICOG and ICMR collection; implement source metadata, versioning, filters, citations, and clinician-controlled attachment.
P4 — Governed pathways and trends	After clinical governance, add configurable trends and clinician-approved administrative pathways with transparent rules and human confirmation.
P5 — High-risk or external integrations	Only after validation and formal review consider prescription/interaction checking, risk stratification, WhatsApp delivery, and lab-report explanation.

Suggested demo sequence

1. Open a synthetic patient and show the timeline.
2. Open Today's Brief and trace an unresolved question to its source.
3. Show task ownership and a handover.
4. Dictate a short note and demonstrate draft review/approval.
5. Open the Clinical Guidelines tab, search a curated source, and show the issuing body, date/version, and original reference.
6. Show how the approved response or summary becomes visible in the patient portal.

8. Safety, privacy and implementation guardrails

MaternaSync is a clinician-first productivity and continuity assistant. The following controls should be treated as product requirements, not optional polish.

Guardrail	Requirement
Human approval	Prescriptions, clinical interpretations, patient-specific recommendations, patient-facing clinical replies, and formal clinical documents must not be released or finalized without authorized professional approval.
No autonomous diagnosis or treatment	Do not position generated summaries, trends, or retrieved guidelines as a diagnosis, treatment recommendation, or autonomous clinical decision.
Transparent sources	Show where information came from, when it was recorded, and whether it is patient-reported, extracted, clinician-entered, or AI-drafted.
Data quality	Display missingness, units, timestamps, and corrections. A missing log is not proof of non-adherence or a clinical problem.
Role-based access	Enforce least-privilege permissions on the server. Protect patient, clinician, and organizational data by role and purpose.
Consent and sharing	Record applicable consent, verify recipients, minimize shared content, and provide revocation/opt-out paths where supported.
Auditability	Record material access, edits, approvals, sharing, task assignment, and consent changes according to policy.
Clinical governance	Clinical pathways, thresholds, educational content, and guideline collections require named ownership, review dates, versioning, and an update process.
Local AI limitations	Local deployment may reduce external data transfer but does not by itself guarantee privacy, security, correctness, or regulatory compliance.
Synthetic data for demos	Use synthetic patient data for development and hackathon demonstrations; do not use real identifiable records without appropriate authorization and safeguards.

India-specific implementation note

The consent, privacy, and health-data architecture should be reviewed against applicable Indian law, institutional requirements, and any relevant ABDM/ABHA integration specifications. Do not claim DPDP or ABHA compliance merely because consent controls or an ABHA-related concept is present.

9. Existing AI healthcare products — inspiration map

The product names below are inspiration references from the supplied planning notes. Capabilities and integrations should be independently verified with official product documentation before implementation or any public comparison.

Product	Inspiration noted	Potential MaternaSync adaptation
Aayi.ai	Between-visit maternal support, patient updates, reminders, trends, concise visit context.	Patient daily log, patient-question intake, reminder workflow, clinician-visible trends after validation.
Nivrit AI	Longitudinal records and AI care-pathway concepts, as described in planning notes.	Continuity Graph, versioned care pathways, administrative task generation after governance.
Gynovate AI	Clinical copilot, documentation and patient education concepts, as described in planning notes.	Patient explanation drafts, document generators, source-attributed clinical reference retrieval.
Ekko MD	AI clinical documentation / ambient-scribe concept, as described in planning notes.	Multilingual voice capture, transcript correction, clinician-approved note drafting.
Vaidhji	Voice-to-SOAP and multilingual documentation concepts, as described in planning notes.	Custom note templates, structured voice notes, candidate ICD-10 coding; external sharing only after privacy review.
Newmi	Regional-language maternal education / companion concept referenced in the proposed feature list.	Clinician-approved trimester education and general patient navigation in supported languages.

Core differentiation

Many individual features — voice transcription, reminders, summaries, and educational content — exist across healthcare products. MaternaSync should differentiate by linking the patient journey to unresolved work, accountable owners, approved communications, handovers, and traceable sources in one continuity layer.

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